FBISE

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2022

CHEMISTRY

Past Paper 2022 • Paper Code: 2SA-I 2208 — D(6082)

Retyped faithfully from the original board paper — same questions, no changes.

Note: CHEMISTRY SSC-II. SECTION – A (Marks 12). Time allowed: 20 Minutes. Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil. Version No. 6082. Fill the relevant bubble against each question:

Q1. Slag is obtained as impurity in the metallurgical processes. Identify slag from the following:

(A) CuO
(B) $FeSiO_3$
(C) CuS
(D) FeO

میٹلرجیکل عمل میں فاضل مواد کے طور پر حاصل کیا جاتا ہے۔ درج شدہ میں سے سلیگ (Slag) کی شناخت کریں:

Q2. Predict the unit of Kc for the given reaction: $N_2(g)+O_2(g) \rightleftharpoons 2NO(g)$, $Kc=[NO]^2/([N_2][O_2])$

(A) No units
(B) $Mol^{-1}dm^{-3}$
(C) $Mol^{-1}\ dm^3$
(D) $Mol\ dm^3$

دیے گئے کیمیائی تعامل کے لیے Kc کے یونٹ کی پیش گوئی کریں:

(A) کوئی اکائی نہیں

Q3. According to Lewis concept which of the following is an electron pair acceptor?

(A) Cl^-
(B) HCl
(C) NaCl
(D) NH_3

لیوس کے نظریے کے مطابق درج شدہ میں سے کون الیکٹرون پیئر قبول کرنے والا ہے؟

Q4. HCl is an acid. Identify the reason from the following:

(A) Can donate proton (H^+)
(B) Contains OH^- group
(C) Can accept proton (H^+)
(D) Can donate an electron pair

HCl ایک ایسڈ ہے۔ درج شدہ میں سے وجہ کی نشاندہی کریں۔

(A) پروٹان (H^+) دے سکتا ہے
(B) OH^- گروپ پر مشتمل ہے
(C) پروٹان (H^+) قبول کر سکتا ہے
(D) الیکٹرون پیئر دے سکتا ہے

Q5. The pH of 10^{-3}M aqueous solution of NaOH is:

(A) 9
(B) 3
(C) 11
(D) 2

NaOH کے $10^{-3}M$ آبی محلول کا pH ____ ہے۔

Q6. Which of the following has general formula R–O–R?

(A) Ester
(B) Alcohol
(C) Aldehyde
(D) Ether

درج شدہ میں سے کس کا عمومی فارمولا R–O–R ہے؟

(A) ایسٹر
(B) الکحل
(C) ایلڈیہائڈ
(D) ایتھر

Q7. Identify the product in the given chemical reaction: H–C(OH)–C(OH)–H —$2H_2O$→ H–C(=O)–C(=O)–H (Note: the structural formulas are given in the source.)

(A) Tetra hydroxy ethane
(B) Oxalic Acid
(C) Glyoxal
(D) Ethyne

دیے گئے کیمیائی تعامل میں پروڈکٹ کی شناخت کریں۔

(A) ٹیٹرا ہائیڈروکسی ایتھین
(B) آکزیلک ایسڈ
(C) گلائی آکسل
(D) ایتھائن

Q8. Unsaturated hydrocarbons have multiple Carbon-Carbon bonds. Which of the following is an unsaturated hydrocarbon?

(A) CH_3–Cl
(B) CH_3–CH_3
(C) CH_4
(D) CH_2=CH–CH_3

غیر سیر شدہ ہائیڈروکاربنز میں ملٹیپل کاربن کاربن بانڈ ہوتے ہیں۔ درج شدہ میں سے کون سا (unsaturated) غیر سیر شدہ ہائیڈروکاربن ہے؟

Q9. Peptide linkage is present in:

(A) Glucose
(B) Vitamins
(C) Carbohydrates
(D) Proteins

پیپٹائڈ لنکیج کس میں موجود ہے؟

(A) گلوکوز
(B) وٹامنز
(C) کاربوہائیڈریٹس
(D) پروٹینز

Q10. Atmosphere is composed of ____ layers.

(A) 5
(B) 2
(C) 3
(D) 4

ماحول (اٹموسفیئر) ____ لیئرز پر مشتمل ہے۔

Q11. Which of the following is a secondary pollutant?

(A) Ammonia
(B) Sulfuric acid
(C) Sulphur dioxide
(D) Carbon dioxide

درج شدہ میں سے کون سا ثانوی پولیوٹنٹ ہے؟

(A) امونیا
(B) سلفیورک ایسڈ
(C) سلفر ڈائی آکسائیڈ
(D) کاربن ڈائی آکسائیڈ

Q12. Slaked lime is used in Solvay process. What is slaked lime?

(A) Sodium hydroxide
(B) Calcium phosphate
(C) Sodium phosphate
(D) Calcium hydroxide

سولوے پراسس میں استعمال ہوتا ہے۔ سلیکڈ لائم (Slaked lame) کیا ہے؟

(A) سوڈیم ہائیڈرو آکسائیڈ
(B) کیلشیم فاسفیٹ
(C) سوڈیم فاسفیٹ
(D) کیلشیم ہائیڈرو آکسائیڈ

Note: SUPPLEMENTARY TABLE (Atomic No / Symbol / Mass No): 1 H 1; 2 He 4; 3 Li 7; 4 Be 9; 5 B 11; 6 C 12; 7 N 14; 8 O 16; 9 F 19; 10 Ne 20; 11 Na 23; 12 Mg 24; 13 Al 27; 14 Si 28; 15 P 31; 16 S 32; 17 Cl 35.5; 18 Ar 40; 19 K 39; 20 Ca 40. —2SA-I 2208-(HA) 6082—. Page 1 of 2. Page 2 of 2.

Note: CHEMISTRY SSC–II. Time allowed: 2:40 Hours. Total Marks Sections B and C: 53. NOTE: Answer any eleven parts from Section 'B' and attempt any two questions from Section 'C' on the separately provided answer book. Write your answers neatly and legibly. (The source also carries the full Urdu translation of this paper.)

Q. SECTION – B (Marks 33)

Q. 2 Answer any ELEVEN parts from the following. All parts carry equal marks. (11 x 3 = 33)

(i) Compare stratosphere and troposphere. (Any three points)

(ii) What is meant by reversible chemical reaction? Describe briefly with an example.

(iii) Briefly explain the Lewis concept of acids and bases with examples.

(iv) Complete and balance following neutralization reactions: (i) $H_2CO_3 + NaOH \rightarrow Na_2CO_3 + H_2O$ (ii) $HCl + Mg \rightarrow MgCl_2 + H_2$ (iii) $NaOH + HCl \rightarrow$ ____ + H_2O.

(v) Water is a weak electrolyte. Justify with the help of self-ionization reaction of water.

(vi) Differentiate between n-propyl and iso-propyl radicals briefly with their structures.

(vii) Write the importance of water briefly.

(viii) Write names of products in following Chemical reactions: a) $CH_4 + Cl_2$ —diffused sunlight→ $CH_3Cl + HCl$ b) $C_2H_4 + Cl–Cl \rightarrow$ 1,2-dichloroethane (structural formulas given in source) c) $CH{\equiv}CH + Cl_2 \rightarrow HC{=}CH$ with Cl atoms below (structural formula given in source).

(ix) Describe one of the following: a) Magnetic separation b) Floatation Process

(x) Describe the origin of Petroleum and Natural Gas

(xi) What are monosaccharides? Write briefly. Also give two examples.

(xii) Acids dissolved in rainwater damage soil and plants. Give reasons

(xiii) What are the causes of hardness in water? List any three.

(xiv) What happens when ammonical brine is passed through carbonation tower. (Solvay Process)

(xv) Describe the composition of Urea.

Q. SECTION – C (Marks 20)

Note: Attempt any TWO questions. All questions carry equal marks. (2 x 10 = 20)

Q. 3 a. State the law of mass action and propose the expression for equilibrium constant for a general reaction. (06) b. What are water borne diseases? Describe any two. (04)

Q. 4 a. What is water pollution? Describe the effects of using polluted water. (05) b. What are vitamins? Write their importance. (05)

Q. 5 a. Make a chart of the alkane, alkene and alkyne functional groups, their general formula and give relevant examples. (06) b. Write the preparation of ethene from alcohol and ethyl bromide. (04)

> Note: —2SA-I 2208 (HA)—